

PRAJNAN KUMAR SARMA

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📍 Guwahati, Assam, India

PROJECTS

Vyapaar Sahayak: Conversational AI Accounting Assistant 🗣️

- Built a voice-enabled AI assistant enabling small shopkeepers to manage invoices, credit, and payments using Hindi/Hinglish natural language commands, addressing a real gap in vernacular AI tools for unbanked micro-businesses
- Designed an end-to-end AI command processing pipeline covering speech-to-text input, intent classification (TF-IDF + Logistic Regression), entity extraction, command routing, and automated accounting operations
- Architected a modular AI command engine decoupling speech processing, NLP inference, and business logic layers; planned upgrade to transformer-based multilingual models (IndicBERT)
- **Tech Stack:** Python, FastAPI, React, SQLite, NLP, TF-IDF, Logistic Regression, Web Speech API

InkSynk: A Smart Real-Time Collaborative Editor 🗣️ |

- Built a real-time collaborative text editor enabling multiple users to edit the same document simultaneously using WebSockets and Yjs (CRDT-based synchronization), ensuring conflict-free updates
- Implemented live user presence indicators, colored cursor tracking, and typing awareness features to enhance real-time collaboration experience
- Designed a full-stack architecture with React frontend, Node.js backend, and MongoDB for document persistence and scalable data management
- Integrated AI-powered document summarization using Hugging Face APIs with a secure Bring Your Own Key (BYOK) approach ensuring user data privacy
- Engineered a backend proxy layer to securely handle third-party API requests and resolve CORS limitations in production deployment

Urban X-Ray: AI Flood Intelligence System(Collaborative) 🗣️

- Developed an AI-based urban flood prediction system modelling drainage health and predicting zone-level flood probability, motivated by recurring waterlogging in rapidly urbanising Indian cities
- Engineered the Drainage Condition Index (DCI), a composite indicator enabling ML-based flood risk classification using Logistic Regression; contributed to a CNN-based flood image detection module for citizen-reported imagery
- **Tech Stack:** Python, Scikit-learn, CNN, Streamlit

Leukemia Cell Detection using ResNet-50: Deep Learning Diagnostic System 🗣️

- Built a deep learning diagnostic system to detect leukemic cells from microscopic blood smear images, targeting resource-constrained pathology settings where manual screening is error-prone and time-intensive
- Achieved 80% classification accuracy using fine-tuned ResNet-50 with automated OpenCV nucleus segmentation and region-of-interest extraction pipeline
- Developed Flask inference backend and Streamlit interface for clinician-facing annotated prediction outputs
- **Tech Stack:** Python, TensorFlow/Keras, ResNet-50, OpenCV, Flask, Streamlit

EcoMitra: Smart Waste Segregation System (Collaborative) 🗣️

- Developed the computer vision module for a hardware-integrated smart waste bin; AI predictions directly controlled the bins mechanical sorting mechanism
- Implemented CNN-based image classification using MobileNetV2 and integrated YOLOv5 for real-time multi-class object detection
- **Tech Stack:** Python, MobileNetV2, YOLOv5, Computer Vision

EXPERIENCE

Google Developer Group on Campus – Gauhati University

PR and Outreach Lead

Sep 2025 – Present

Guwahati, India

- Led outreach initiatives increasing participation in technical events and hackathons
- Organized developer meetups, workshops, and collaborations

Expelee

Research Intern

May 2025 – Jul 2025

Remote

- Conducted market and competitor research for emerging tech platforms
- Supported product strategy through data-driven insights

ACHIEVEMENTS

Secured **3rd Place at GUenARK 2026**, a state-level hackathon with participation from multiple institutions

Secured **3rd Place at IIIT Guwahati E-Summit Inizio 2026**, a national-level hackathon and innovation competition

EDUCATION

Gauhati University

Bachelor of Technology in Information Technology

Jul 2024 – Aug 2028

Guwahati, Assam

RELEVANT COURSEWORK

Data Structures & Algorithms • OOP • DBMS • Computer Networks • Operating Systems • Machine Learning • Linear Algebra

SKILLS

Languages: Python, C, C++, JavaScript

AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision

Frameworks: TensorFlow, Pytorch, Scikit-learn, React, Flask, FastAPI

Tools: Git, GitHub, SQLite, VS Code

Design: Figma, Canva